ATM Error Resolution Guide

Comprehensive Troubleshooting Manual

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Card Reader Errors

Error Code: CR-001 - Card Not Ejected / Card Jam

Priority: Critical

Symptoms:

- Card stuck in reader
- "Card retained" message displayed
- Reader motor not responding

Resolution Steps:

1. Press the emergency card eject button (usually located inside service panel)
2. If manual eject fails, power cycle the card reader module
3. Open the card reader compartment and check for:
    - Physical obstructions
    - Damaged or worn cards stuck in path
    - Debris or foreign objects
4. Inspect the card transport rollers for wear or damage
5. Clean the magnetic stripe reader head with isopropyl alcohol
6. Test with a known good card
7. If issue persists, replace the card reader module
8. Update card reader firmware if available

Prevention:

- Regular cleaning schedule (weekly)
- Inspect cards for damage before insertion
- Replace worn rollers every 6 months

Error Code: CR-002 - Card Read Error

Priority: High

Symptoms:

- "Unable to read card" message
- Multiple swipe attempts required
- Intermittent read failures

Resolution Steps:

1. Clean the magnetic stripe reader head with approved cleaning card
2. Verify card is not damaged or demagnetized
3. Check for proper card insertion (correct orientation)
4. Inspect reader head for physical damage
5. Test with multiple cards to isolate issue
6. Check reader connection cables
7. Verify reader voltage levels (should be 12V ±0.5V)
8. Replace reader head if cleaning doesn't resolve issue
9. Recalibrate the reader sensor

Error Code: CR-003 - Chip Card Read Failure

Priority: High

Symptoms:

- EMV chip cannot be read
- "Insert card again" prompts
- Fallback to magnetic stripe

Resolution Steps:

1. Clean chip reader contacts with compressed air
2. Verify chip reader pins are not bent or damaged
3. Test with multiple chip cards
4. Check chip reader firmware version
5. Update to latest EMV certification
6. Verify chip reader power supply (3.3V)
7. Replace chip reader module if necessary
8. Recertify EMV compliance after replacement

Cash Dispenser Errors

Error Code: CD-001 - Cash Dispenser Jam

Priority: Critical

Symptoms:

- Notes stuck in dispenser path
- Partial dispensing
- "Dispenser failure" error
- Transaction incomplete

Resolution Steps:

1. Put ATM in supervisor mode
2. Access cash dispenser module
3. Open dispenser door using proper key
4. Remove jammed notes carefully (note serial numbers for reconciliation)
5. Inspect the bill path for:
    - Torn or damaged notes
    - Foreign objects
    - Dust accumulation
6. Check pick rollers for wear (replace if rubber is hard or cracked)
7. Inspect separation rollers
8. Clean all rollers with lint-free cloth
9. Verify cassette is properly seated
10. Check note thickness sensor
11. Test dispense with small amount
12. Update dispenser error log
13. Reconcile cash and update journal

Prevention:

- Load only new, crisp notes
- Regular roller cleaning (daily)
- Replace pick rollers every 3 months
- Maintain proper humidity levels (40-60%)

Error Code: CD-002 - Cassette Empty

Priority: Medium

Symptoms:

- "Out of service" message
- Cannot complete withdrawals
- Low cash warning ignored

Resolution Steps:

1. Verify cassette is actually empty (physical count)
2. Check cassette sensor functionality
3. Replenish cassette with proper denomination
4. Reset cassette counter in ATM software
5. Verify note count matches system count
6. Test small dispense to confirm functionality
7. Update cash management system
8. Clear "out of service" status

Error Code: CD-003 - Double Note Detection

Priority: High

Symptoms:

- Transaction rejected
- "Dispenser error" message
- Notes dispensed incorrectly

Resolution Steps:

1. Remove cassette and inspect notes
2. Check for stuck-together notes
3. Verify notes are same denomination and condition
4. Clean double-detect sensors
5. Adjust pick roller pressure if adjustable
6. Check for humidity issues causing notes to stick
7. Replace damaged or worn notes
8. Calibrate double-detect sensors
9. Test with various note conditions

Error Code: CD-004 - Cassette Not Recognized

Priority: High

Symptoms:

- "Cassette error" message
- Dispenser won't initialize
- Wrong denomination detected

Resolution Steps:

1. Remove and reseat cassette
2. Check cassette connector pins for damage
3. Verify correct cassette type for ATM model
4. Check cassette lock mechanism
5. Inspect cassette sensor
6. Verify cassette configuration in software
7. Clean cassette connector
8. Replace cassette if damaged
9. Reconfigure in ATM settings

Network & Communication Errors

Error Code: NET-001 - Network Timeout

Priority: High

Symptoms:

- "Unable to connect" message
- Transaction timeouts
- Slow response times

Resolution Steps:

1. Check ethernet cable connections (both ends)
2. Verify link lights on network port (should be green/blinking)
3. Ping gateway: `ping [gateway_ip]`
4. Ping DNS server: `ping 8.8.8.8`
5. Check router/switch status
6. Verify DHCP lease is active
7. Review firewall rules
8. Check network congestion
9. Test with alternate network cable
10. Restart network interface
11. Contact ISP if external connectivity issue
12. Review and increase timeout settings if needed

Logs to Check:

- Network interface logs
- Transaction logs
- Router/switch logs

Error Code: NET-002 - Host Connection Lost

Priority: Critical

Symptoms:

- Cannot authorize transactions
- "System unavailable" message
- Frequent disconnections

Resolution Steps:

1. Verify host IP address and port
2. Check VPN connection status
3. Test connectivity: `telnet [host_ip] [port]`
4. Review SSL/TLS certificates (check expiration)
5. Verify firewall allows ATM traffic
6. Check for DNS resolution issues
7. Review host system status
8. Inspect encryption keys
9. Check for IP conflicts
10. Contact host support team
11. Review connection logs for patterns
12. Test backup host connection if available

Error Code: NET-003 - SSL/TLS Certificate Error

Priority: Critical

Symptoms:

- Certificate validation failures
- Secure connection refused
- Authentication errors

Resolution Steps:

1. Check certificate expiration date
2. Verify certificate chain is complete
3. Update root certificates
4. Check system time/date (certificate validation depends on accurate time)
5. Verify certificate hostname matches
6. Review certificate revocation list (CRL)
7. Check OCSP responder status
8. Reinstall certificates if corrupted
9. Contact certificate authority if needed
10. Update certificate store

Printer Errors

Error Code: PR-001 - Receipt Printer Jam

Priority: Medium

Symptoms:

- Paper stuck in printer
- Partial receipt printing
- "Printer error" message

Resolution Steps:

1. Open printer cover
2. Remove jammed paper carefully
3. Check paper path for:
    - Torn paper pieces
    - Foreign objects
    - Dust buildup
4. Clean print head with isopropyl alcohol
5. Clean paper sensors
6. Inspect paper feed rollers
7. Verify paper roll is loaded correctly
8. Check paper quality (thermal paper appropriate for printer)
9. Adjust paper tension if adjustable
10. Replace paper if damaged or poor quality
11. Test print small sample
12. Clean cutter blade if equipped

Prevention:

- Use quality thermal paper
- Regular cleaning (weekly)
- Replace paper rolls before complete depletion

Error Code: PR-002 - Printer Out of Paper

Priority: Low

Symptoms:

- "No receipt available" message
- Empty paper roll
- Paper sensor triggered

Resolution Steps:

1. Replace paper roll with new roll
2. Ensure paper feeds correctly
3. Check paper sensor is not blocked
4. Verify paper roll orientation (thermal side down)
5. Test print functionality
6. Update maintenance log
7. Check paper inventory

Error Code: PR-003 - Print Head Overheating

Priority: High

Symptoms:

- Faded printing
- Intermittent printing
- "Printer temperature error"

Resolution Steps:

1. Allow printer to cool down (15-20 minutes)
2. Check printer ventilation
3. Verify print head heat sink is clean
4. Reduce print density settings
5. Check for excessive printing cycles
6. Inspect cooling fan operation
7. Clean print head properly
8. Replace thermal paper with quality paper
9. Check ambient temperature (should be below 30°C)
10. Replace print head if damaged

Display & Touch Screen Errors

Error Code: DS-001 - Touch Screen Unresponsive

Priority: Critical

Symptoms:

- Screen not responding to touch
- Intermittent touch response
- Calibration issues

Resolution Steps:

1. Restart touch screen driver from service menu
2. Check USB connection between screen and PC
3. Verify power supply to screen (12V/24V depending on model)
4. Clean screen surface with approved cleaner
5. Check for physical damage to screen
6. Run touch screen calibration utility
7. Test with stylus/finger to isolate issue
8. Check touch controller board connections
9. Update touch screen drivers
10. Replace touch overlay if damaged
11. Replace touch controller if faulty

Error Code: DS-002 - Display Flickering

Priority: Medium

Symptoms:

- Screen flickers or dims
- Intermittent display
- Brightness fluctuations

Resolution Steps:

1. Check video cable connections
2. Verify LCD backlight power supply
3. Adjust brightness and contrast settings
4. Check for electromagnetic interference
5. Inspect video cable for damage
6. Test with alternate video cable
7. Check graphics driver settings
8. Verify power supply voltage stability
9. Replace LCD inverter if equipped
10. Replace LCD panel if backlight failure

Error Code: DS-003 - Screen Blank/No Display

Priority: Critical

Symptoms:

- Black screen
- No backlight
- System running but no video

Resolution Steps:

1. Check monitor power switch
2. Verify power cable connections
3. Check brightness setting (may be turned down)
4. Test video signal with external monitor
5. Check video cable connection
6. Verify PC is booting (listen for sounds)
7. Check LCD power supply
8. Test backlight inverter
9. Replace LCD if no backlight
10. Check graphics card/onboard video
11. Boot into safe mode to test

PIN Pad Errors

Error Code: PP-001 - PIN Pad Malfunction

Priority: Critical

Symptoms:

- Keys not responding
- Incorrect key presses registered
- EPP error messages

Resolution Steps:

1. Test each key individually
2. Check PIN pad connection cable
3. Verify PIN pad power supply
4. Check for physical damage to keypad
5. Clean keypad surface and contacts
6. Test EPP (Encrypted PIN Pad) mode
7. Verify encryption keys are loaded
8. Check PIN pad firmware version
9. Reload encryption keys if needed
10. Replace PIN pad if hardware failure
11. Recertify PCI compliance after replacement

Security Notes:

- Log all PIN pad errors for PCI compliance
- Report suspicious behavior immediately
- Follow tamper-evident seal procedures

Error Code: PP-002 - Encryption Key Error

Priority: Critical

Symptoms:

- "PIN pad not secure" message
- Transaction declines
- Key injection failure

Resolution Steps:

1. Verify master key is loaded
2. Check key expiration dates
3. Reload working keys from secure source
4. Verify key serial numbers
5. Test with key management system
6. Check for tamper detection triggers
7. Verify PIN pad certificate
8. Contact security team for key reload
9. Document all key loading procedures
10. Test PIN encryption after reload

System & Software Errors

Error Code: SYS-001 - Operating System Failure

Priority: Critical

Symptoms:

- Boot failure
- Blue screen of death (Windows)
- System freeze

Resolution Steps:

1. Attempt safe mode boot
2. Check boot drive connections
3. Run system diagnostics
4. Check hard drive health (SMART status)
5. Verify BIOS settings
6. Run disk check utility
7. Restore from last known good configuration
8. Check system logs for errors
9. Restore from backup image if available
10. Replace hard drive if failed
11. Reinstall operating system as last resort
12. Restore ATM software and configuration

Error Code: SYS-002 - Application Crash

Priority: High

Symptoms:

- ATM software unexpectedly closes
- Error messages
- Automatic restart

Resolution Steps:

1. Check application logs
2. Verify all services are running
3. Check for software updates
4. Verify configuration files
5. Check disk space (should have >20% free)
6. Review memory usage
7. Scan for malware
8. Restore application from backup
9. Check database connectivity
10. Restart application services
11. Contact software vendor if persistent

Error Code: SYS-003 - Database Error

Priority: High

Symptoms:

- Transaction history unavailable
- Configuration errors
- Data synchronization failures

Resolution Steps:

1. Check database service status
2. Verify database connection string
3. Check database disk space
4. Run database integrity check
5. Review database logs
6. Test database connectivity
7. Backup database before repairs
8. Run database repair utility
9. Restore from backup if corrupted
10. Optimize database tables
11. Update database drivers

Deposit Module Errors

Error Code: DM-001 - Envelope Jam

Priority: High

Symptoms:

- Envelope stuck in acceptor
- Deposit incomplete
- Module not accepting envelopes

Resolution Steps:

1. Enter supervisor mode
2. Open deposit module
3. Remove jammed envelope
4. Document envelope number and customer details
5. Check deposit path for obstructions
6. Inspect feed rollers
7. Clean rollers and sensors
8. Test with standard envelope
9. Verify envelope meets specifications
10. Update transaction log

Error Code: DM-002 - Check Image Capture Failure

Priority: High

Symptoms:

- Poor image quality
- Check not recognized
- Image processing error

Resolution Steps:

1. Clean scanner glass/camera lens
2. Verify check imaging camera alignment
3. Check lighting (LED array)
4. Adjust image capture settings
5. Test with standard check
6. Calibrate image sensors
7. Update image processing software
8. Check camera connection
9. Replace camera module if faulty
10. Verify image storage capacity

Environmental & Hardware Errors

Error Code: ENV-001 - Temperature Warning

Priority: Medium

Symptoms:

- High temperature alert
- System performance degradation
- Automatic shutdown

Resolution Steps:

1. Check ambient temperature (should be 10-40°C)
2. Verify air conditioning is functioning
3. Check cooling fans operation
4. Clean air filters
5. Check for blocked ventilation
6. Inspect heat sinks for dust
7. Monitor CPU temperature
8. Check for excessive sunlight exposure
9. Verify proper cabinet insulation
10. Add additional cooling if needed

Error Code: ENV-002 - Power Supply Failure

Priority: Critical

Symptoms:

- Unexpected shutdowns
- Voltage warnings
- Components not powering on

Resolution Steps:

1. Check main power connection
2. Verify UPS battery status
3. Test voltage output (should be 12V, 5V, 3.3V as required)
4. Check for power surges/sags
5. Inspect power cables for damage
6. Test with multimeter
7. Replace fuses if blown
8. Check circuit breaker
9. Replace power supply unit if faulty
10. Test all components after replacement

Error Code: ENV-003 - UPS Battery Low

Priority: High

Symptoms:

- Battery warning messages
- Short backup time
- Frequent battery alarms

Resolution Steps:

1. Check UPS battery age (replace every 3-5 years)
2. Test battery capacity
3. Verify battery connections
4. Check charging circuit
5. Measure battery voltage
6. Replace battery if below specification
7. Test UPS under load
8. Update UPS firmware
9. Configure proper shutdown procedures
10. Document battery replacement

Emergency Procedures

Complete ATM Shutdown

1. Put ATM in "Out of Service" mode
2. Complete all pending transactions
3. Secure all cash cassettes
4. Lock card reader
5. Shutdown software application
6. Power down PC
7. Turn off peripheral devices
8. Disconnect main power
9. Lock ATM cabinet
10. Place "Out of Order" sign
11. Document reason and time

Transaction Reversal

1. Document transaction details
2. Verify customer complaint
3. Check transaction logs
4. Review video surveillance if available
5. Contact host for transaction status
6. Follow bank's reversal procedures
7. Document all steps taken
8. Notify customer of resolution

Security Breach Response

1. Immediately disconnect ATM from network
2. Secure physical location
3. Preserve all logs and evidence
4. Contact security team
5. Document all suspicious activity
6. Review surveillance footage
7. Inspect for tampering devices
8. Follow incident response plan
9. Notify law enforcement if required
10. Conduct forensic analysis

Preventive Maintenance Schedule

Daily Tasks

- Visual inspection of ATM condition
- Check cash levels
- Verify receipt paper supply
- Clean screen and keypad
- Review transaction logs

Weekly Tasks

- Clean card reader
- Clean cash dispenser rollers
- Test all functions
- Check error logs
- Clean printer

Monthly Tasks

- Full system diagnostics
- Update software/firmware
- Replace worn components
- Deep clean all modules
- Battery backup test
- Security seal inspection

Quarterly Tasks

- Replace pick rollers
- Comprehensive system test
- Update virus definitions
- Review and update configurations
- Train staff on new procedures

Annual Tasks

- Replace UPS batteries
- Complete hardware audit
- Full system backup
- Security assessment
- PCI compliance review
- Contract service review

Contact Information

Technical Support Hotline: 1-800-ATM-HELP
Email: atm.support@bank.com
Emergency Line: 1-800-ATM-911 (24/7)

Escalation Levels:

- Level 1: Local technician
- Level 2: Regional support center
- Level 3: Manufacturer technical support
- Level 4: Engineering team

Revision History

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1.0	2026-01-05	Initial document creation	ATM Support Team

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